

MATHEMATICS FOR ML: DERIVATIVES & OPTIMIZATION

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Outline

1 DERIVATIVES & OPTIMIZATION

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- ▶ Feel free to follow me at:
 - ▶ Github (github.com/yogeshhk)
 - ▶ LinkedIn (www.linkedin.com/in/yogeshkulkarni/)
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Office Hours:
Saturdays, 2 to 5pm
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About Content

- ▶ Bare essential mathematics needed for Machine/Deep Learning
- ▶ Just to get you started.
- ▶ Each field within Machine/Deep Learning can go extremely deep.
- ▶ This is to give you enough foundation needed.

About Me

- ▶ I am not a mathematician.
- ▶ And this course won't turn YOU into one!!! (if you are not already)
- ▶ But, I will surely attempt to make you get interested in the learning, more.

Differentiation

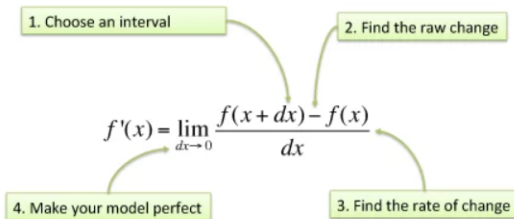
Background Theories

- ▶ **Limits:** We let the measurement artifacts get smaller and smaller until they effectively disappear (cannot be distinguished from zero).
- ▶ **Infinitesimals:** Create a new type of number that lets us try infinitely-small change on a separate, tiny number system. When we bring the result back to our regular number system, the artificial elements are removed.

(Ref: Lesson 10: The Theory Of Derivatives - Better Explained)

The Formula

The Derivative



(Ref: Lesson 10: The Theory Of Derivatives - Better Explained)

The Steps

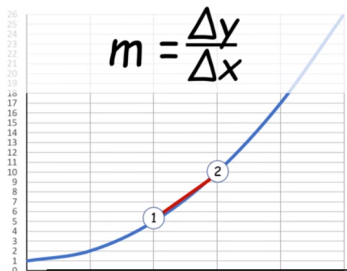
Step	Example
Start with function to study	$f(x) = x^2$
1. Increase the input by dx , a sample change	$f(x + dx) = (x + dx)^2 = x^2 + 2x \cdot dx + (dx)^2$
2. Find the resulting increase in output, df	$df = f(x + dx) - f(x) = 2x \cdot dx + (dx)^2$
3. Find the ratio of output change to input change	$\frac{df}{dx} = \frac{2x \cdot dx + (dx)^2}{dx} = 2x + dx$
4. Throw away any measurement artifacts	$2x + dx \xrightarrow{dx=0} 2x$

(Ref: Lesson 10: The Theory Of Derivatives - Better Explained)

Derivative

Slope of a graph between a section (ie two points) can be easily calculated.

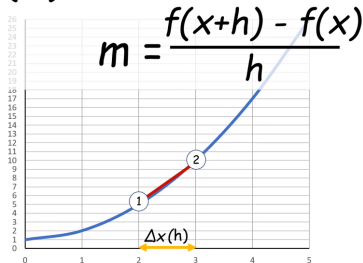
$$f(x) = x^2 + 1$$



Derivative

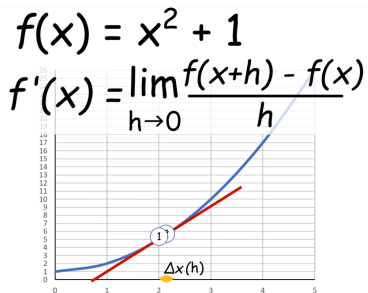
Delta x can be replaced by h and the formula for segment slope becomes.

$$f(x) = x^2 + 1$$



Derivative

If both points come very close then we can say that the 'Segment Slope' becomes a 'Point Slope'.



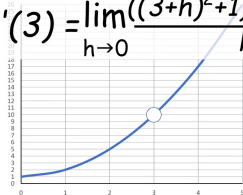
The concept of Limits is used to determine RATE OF CHANGE at a given POINT.

Derivative

How to calculate value of the derivative at a points, say $x = 3$?

$$f(x) = x^2 + 1$$

$$f'(3) = \lim_{h \rightarrow 0} \frac{((3+h)^2 + 1) - (3^2 + 1)}{h}$$



$$f'(3) = 6$$

Derivative

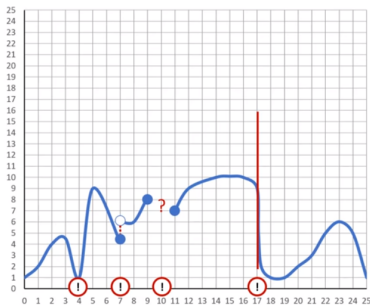
- ▶ Slope at a point on curve is linear, where as the curve itself is not linear.
- ▶ Tangent plane at a point on a sphere is linear but the sphere is not.
- ▶ So, derivative is a Linear approximation

Differentiability

- ▶ Note: A function may not be differentiable at every point; that is you might not be able to calculate the derivative for every point on the function line.
- ▶ To be differentiable at a given point, the function must be continuous at that point,
- ▶ The tangent line at that point cannot be vertical, and the line must be smooth at that point
- ▶ Cannot suddenly change direction or have kinks

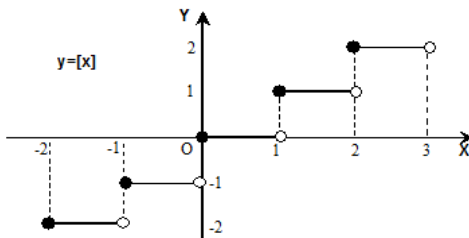
Continuous Functions

- ▶ A function below is not defined at certain points ($x = 10$)
- ▶ At certain point ($x = 7$), it is multi valued.
- ▶ At certain point ($x = 17$), the slope is vertical, so not differentiable.
- ▶ At certain point ($x = 4$), there is sudden change in direction, so not differentiable.



Discontinuous Functions

Step function: $f(x) = [x]$ integer part of x . This suddenly jumps from 0 to 1 to 2, etc.



Domain $\rightarrow \mathbb{R}$;

Period $\rightarrow$ non periodic;

Range $\rightarrow \mathbb{I}$;

Nature $\rightarrow$ neither even nor odd

Differentiability

- ▶ $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} = A$ if limit exists.
- ▶ In the above formula x_0 is constant, so this function is wrt h
- ▶ So, $\lim_{h \rightarrow 0} g(h) = A$
- ▶ For any given arbitrary $\epsilon > 0$, there exists $\delta > 0$, so that if $0 < |h| < \delta$, then $|g(h) - A| < \epsilon$
- ▶ However small ϵ can be made, we can still find corresponding δ . The dependence is on f as well as the point at which we are calculating.

Example

- ▶ $f(x) = x^2$, at $x_0 = 1$
- ▶ $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = 2x + h$
- ▶ Prove: $(x^n)' = nx^{n-1}$

$$(x^n)' = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \quad (1)$$

$$= \lim_{h \rightarrow 0} \frac{x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + h^n - x^n}{h} \quad (2)$$

$$= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + h^{n-1} \right] \quad (3)$$

$$\text{▶ } \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + h^{n-1} \right] = nx^{n-1}$$

Example

- ▶ $f(x) = x^2 + x$, at $x_0 = 5$
- ▶ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- ▶ $f'(x) = \lim_{h \rightarrow 0} \frac{((x+h)^2 + x+h) - (x^2 + x)}{h}$
- ▶ $f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + h^2 + 2xh + x + h - x^2 - x}{h}$
- ▶ $f'(x) = \lim_{h \rightarrow 0} 2x + h + 1$
- ▶ Evaluating the limit, ie $h = 0$, becomes $f'(x) = 2x + 0 + 1$
- ▶ For $x = 5$ the result is $f'(5) = 2 \cdot 5 + 1 = 10 + 1 = 11$

Differentiation formulas

Let $f = f(x)$ and $g = g(x)$ have derivatives at $x = a$. Then

- ▶ $(f + g)'(a) = f'(a) + g'(a)$
- ▶ $(f - g)'(a) = f'(a) - g'(a)$
- ▶ $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$
- ▶ $\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{[g(a)]^2}$

In general, the derivative $f'(a)$ may be used by other notations:

$$Df(a) \quad \frac{d}{dx}f(a) \quad \frac{df}{dx}(a) \quad \frac{df}{dx} \Big|_{x=a} \quad Df \Big|_{x=a}$$

Theorems

Let f and g be differentiable on I , then so are the following functions.
Moreover,

- ▶ $(f + g)' = f' + g'$
- ▶ $(f - g)' = f' - g'$
- ▶ $(fg)' = f'g + fg'$
- ▶ $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

Concept	Key Analogy / Takeaway
Multiplication Rule $(f \cdot g)' = f \cdot dg + g \cdot df$	Grow a garden on two sides; ignore the corner. 

(Ref: Lesson 4: Learning The Official Terms - Better Explained)

The chain rule

Let $g = g(x)$ be a differentiable function at a , and $f = f(x)$ differentiable at $g(a)$. Then the composite function $f \circ g$ is differentiable at a , and

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

In Leibniz notation, let $y = f(u)$ and $u = g(x)$ be both differentiable, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

- ▶ Find $f'(x)$ if $f(x) = \sqrt{x^2 + 1}$.
- ▶ Differentiate (a) $y = \sin(x^2)$ (b) $y = \sin^2 x$.
- ▶ Differentiate $f'(x)$ if $f(x) = (x^3 - 1)^{100}$.
- ▶ Find $f'(x)$ if $f(x) = \frac{1}{\sqrt[3]{x^2 + x + 1}}$.
- ▶ Find the derivative of $g(t) = \left(\frac{t-2}{2t+1}\right)^9$.
4[pt]
- ▶ Differentiate $y = (2x + 1)^5(x^3 - x + 1)^4$.
- ▶ Find $f'(x)$ if $f(x) = \sin(\cos(\tan x))$.
- ▶ Differentiate $y = \sqrt{\sec x^3}$.

2nd order Derivative

2nd order differentiation

- ▶ Finding rate of change of rate of change !!
- ▶ Velocity is rate of change of Distance
- ▶ Whats rate of change of Velocity?

Need 2nd order derivative in finding type of point where derivative goes 0, ie maxima or minima.

Maxima or Minima

$$\begin{aligned}f(x) &= x^2 \\f'(x) &= 2x \\x &= 0 \\f''(x) &= 2\end{aligned}$$



The derivative itself is a function. Its 2nd order derivative is coming to be a constant $+2$. So first order derivative is crossing from negative to positive. Thus its a Minima point.

Maxima or Minima

$$f(x) = -x^2 + 1$$

$$f'(x) = -2x$$

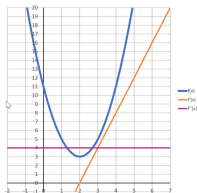
$$f''(x) = -2$$



Its 2nd order derivative is coming to be a constant -2 . So first order derivative is crossing from positive to negative. Thus its a Maxima point.

Optimization

$$\begin{aligned}f(x) &= 2(x-2)^2 + 3 \\&= 2x^2 - 8x + 11 \\f'(x) &= 4x - 8 \\x &= 2 \\f''(2) &= 4\end{aligned}$$



First derivative goes to 0 at $x = 2$. At that point second derivative is positive. Thus its a Minima point.

Partial Derivative

Partial Derivative

- Functions can be of more-than-one-variable, say two variables: x and y .
- When plotted it will be 3D plot. 2 input variables and 1 output variable, total 3.
- For finding derivatives of such functions: we do it turn by turn.
- First wrt x then wrt y and so on. While taking derivative wrt x , the y is considered as constant.

$$\begin{aligned}
 f(x, y) &= x^2 + y^2 \\
 \frac{d(x, y)}{dx} &= \frac{d(x^2 + y^2)}{dx} & \frac{d(x, y)}{dy} &= \frac{d(x^2 + y^2)}{dy} \\
 \frac{dx^2}{dx} &= 2x & \frac{dy^2}{dx} &= 0 & \frac{dx^2}{dy} &= 0 & \frac{dy^2}{dy} &= 2y \\
 \frac{d(x, y)}{dx} &= 2x & \frac{d(x, y)}{dy} &= 2y
 \end{aligned}$$

Gradient

1. Partial derivatives are important if you want to find the analog of the slope for multi-dimensional surfaces. We call this quantity the gradient.
2. For $f(x, y) = x^2 + y^2$ the partial derivatives are:

$$\frac{\partial f(x, y)}{\partial x} = 2x \quad (4)$$

$$\frac{\partial f(x, y)}{\partial y} = 2y \quad (5)$$

3. The Gradient

$$\text{grad}(f(x, y)) = \vec{g}(x, y) = \begin{bmatrix} \frac{\partial f(x, y)}{\partial x} \\ \frac{\partial f(x, y)}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

Plotting the Gradient

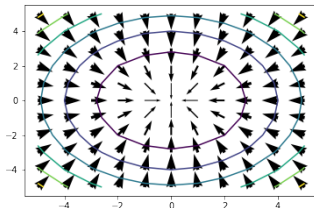
```
1 import matplotlib.pyplot as plt
2 import numpy as np
3 import math
4 ## Create a uniform grid
5 el = np.arange(-5,6)
6 nx, ny = np.meshgrid(el, el, sparse=False, indexing='ij')
7 ## flatten the grid to 1-d and compute the value of the function z
8 x_coord = []
9 y_coord = []
10 z = []
11 for i in range(11):
12     for j in range(11):
13         x_coord.append(float(-nx[i,j]))
14         y_coord.append(float(-ny[i,j]))
15         z.append(nx[i,j]**2 + ny[i,j]**2)
16 x_grad = [-2 * x for x in x_coord]
17 y_grad = [-2 * y for y in y_coord]
```

Plotting the Gradient

```
1  ## Plot the arrows using width for gradient
   plt.xlim(-5.5,5.5)
3  plt.ylim(-5.5,5.5)
   for x, y, xg, yg in zip(list(x_coord), list(y_coord), list(x_grad),
                           list(y_grad)):
5       if x != 0.0 or y != 0.0: ## Avoid the zero divide when scaling the arrow
           l = math.sqrt(xg**2 + yg**2)/2.0
7           plt.quiver(x, y, xg, yg, width = l, units = 'dots')

9  ## Plot the countours of the function surface
   z = np.array(z).reshape(11,11)
11 plt.contour(e1, e1, z)
```

Plotting the Gradient



- ▶ The arrows in the plot point in the direction of the gradient.
- ▶ The width of the arrows is proportional to the value of the gradient. The width of the arrows and the gradient decreases as function gets closer to the minimum. If this is the case everywhere, you can say that a function is convex. It is always much easier to find minimum of convex functions.
- ▶ The direction of the gradient is always perpendicular to the contours.

Optimization

Optimization

- ▶ Goal of Machine Learning is Optimization
- ▶ Trying to find the “best” model for prediction
- ▶ The “best” means that which minimizes prediction error.
- ▶ The technique, widely used is called as “Gradient Descent”

Example

Trying to find relation (formula/model) between input x and output y

x	y
2	4
6	12
3	7
4	7
12	24
21	40

Any guesses?

Example

How do you even start?

Assume $y = w.x$

x	y
2	4
6	12
3	7
4	7
12	24
21	40

Now, how to find w ?

Example

Assume random value, say $w = 1$.

Using the relation, predict y , called y'

x	y	y'
2	4	2
6	12	6
3	7	3
4	7	4
12	24	12
21	40	21

Is y' matching actual y ? NO. What to do? Need to find such w for which both match, right?

Example

How much OFF we are?

x	y	y'	$y - y'$
2	4	2	2
6	12	6	6
3	7	3	4
4	7	4	3
12	24	12	12
21	40	21	19

Too much of diff. Summation of squares (to nullify effect of sign) will give total error. Huge!!

Lets try another value for w and see if that works.

Example

Lets try $w = 0.5$.

Using the relation, and new w predict y find the error again

x	y	y'	$y - y'$
2	4		
6	12		
3	7		
4	7		
12	24		
21	40		

Error going up or down? UP too much. So out “direction” of change was not good. Instead of reducing w to 0.5, we should try increasing it to 1.5.

Example

Lets try $w = 1.5$.

Using the relation, and new w predict y find the error again

x	y	y'	$y - y'$
2	4		
6	12		
3	7		
4	7		
12	24		
21	40		

Error going up or down? Better. So out “direction” of change good but still error does not seem to close to 0. Increase w further.

Example

Lets try $w = 2$.

Using the relation, and new w predict y find the error again

x	y	y'	$y - y'$
2	4		
6	12		
3	7		
4	7		
12	24		
21	40		

Looks far better. Best? Probably. So relation now is $y = 2.x$. Seems to be correct for all rows except 1 or 2. But thats ok.

Maximization Minimization

- ▶ Ideas is to minimize the differences, also called as Error, Loss, Cost.
- ▶ It depends on w
- ▶ So, this is a minimization problem of Cost wrt w
- ▶ How do you find minimum of ANY function?

Maximization Minimization

- ▶ Suppose we have a function f taking vector (or list of values) and outputs a single number.
- ▶ We frequently need to maximize or minimize the the function.
- ▶ Meaning, we wish to try many input vectors and find one where the result is minimum or maximum
- ▶ Examples: minimize cost of operations, maximize profits, etc.

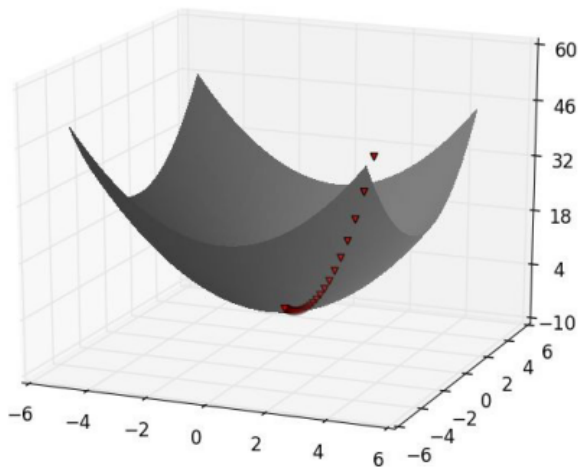
Example

- ▶ Let's use gradients to find the minimum among all three-dimensional vectors
- ▶ We'll just pick a random starting point
- ▶ Take tiny steps in the opposite direction of the gradient
- ▶ Until we reach a point where the gradient is very small.

Say, our function is $\sum x_i^2$

```
1 def sum_of_squares(v):  
    """computes the sum of squared elements in v"""  
3     return sum(v_i**2 for v_i in v)
```


Example



Example

Random vector, starting point

```
1 v = [random.randint(-10,10) for i in range(3)]  
3 # Our Gradient function is, derivative of  $\sum x_i^2$  which is  $\sum 2 x_i$   
5 def sum_of_squares_gradient(v):  
    return [2 * v_i for v_i in v]
```

Example

Step is computed as below, if size is -ve then its in opposite of gradient.

```
2 def step(v, direction, step_size):  
    """move step_size in the direction from v"""  
    return [v_i - step_size * direction_i for v_i, direction_i in zip(v,  
        direction)]
```

Example

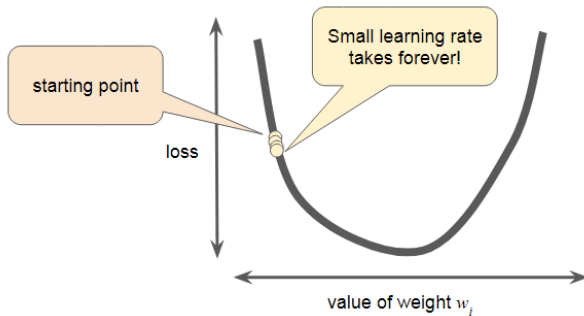
Core logic: We end-up at $[0, 0, 0]$ which is obvious.

```
1 tolerance = 0.0000001
  while True:
3     gradient = sum_of_squares_gradient(v)
      next_v = step(v, gradient, 0.01)
5     if gradient < tolerance:
        break
7     v = next_v
```

Considerations

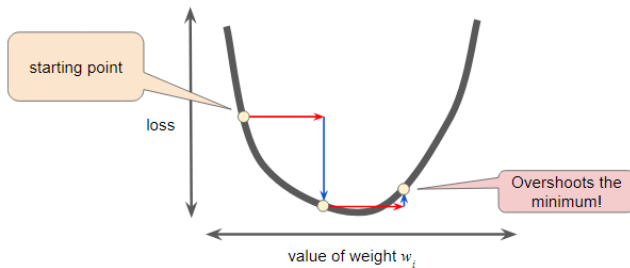
- ▶ Choosing the right step size is critical
- ▶ Using fixed step size
- ▶ Gradually shrinking step size

Learning rate is too small



(Reference: <https://developers.google.com/machine-learning/crash-course/reducing-loss/learning-rate>)

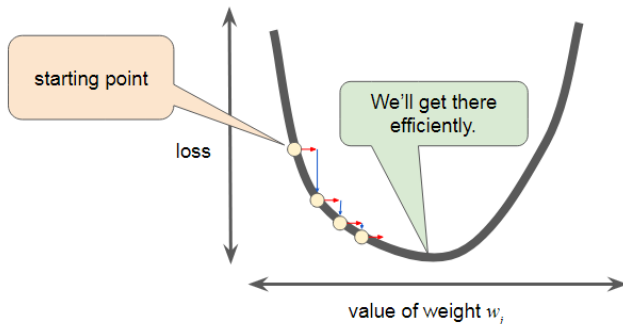
Learning rate is too large



(Reference: <https://developers.google.com/machine-learning/crash-course/reducing-loss/learning-rate>)

Learning rate is just right

The Goldilocks value is related to how flat the loss function is. If you know the gradient of the loss function is small then you can safely try a larger learning rate, which compensates for the small gradient and results in a larger step size.



(Reference: <https://developers.google.com/machine-learning/crash-course/reducing-loss/learning-rate>)

Calculus: Conclusion

We saw ...

- ▶ Calculus deals with infinitesimal processes.
- ▶ Limiting situations.
- ▶ Derivatives
- ▶ Gradients
- ▶ Optimization

Archimedes ...

- ▶ Archimedes had a calculus mindset
- ▶ He re-arranged discs, cylinders, cones, etc. to make “easy to measure” slices
- ▶ As you are doing the same here ...
- ▶ This will make him tear up!!

(Ref: Calculus, Better Explained: Summary - Better Explained)

Thanks ...

- ▶ Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
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